

Earning a Dynamics Claim: An Execution-Grounded Admission Audit for Robot Action Verification

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Abstract—Dynamic robot-action benchmarks should earn a dynamics interpretation: high verification accuracy may instead arise from current geometry, action templates, unordered motion, or a short history suffix. We formulate benchmark admission as an execution-grounded intervention with three gates. Integrity ties labels to candidate executions under a declared observable contract; temporal necessity progressively removes named shortcut routes; and method headroom requires comparison to the strongest fair analytic or geometric control. Our controlled state-observation identification test uses 92,000 GPU-PhysX candidate executions, matched final observations and final-two-frame suffixes, action-diverse sets, and three independent out-of-distribution axes. A lightweight GRU obtains 1.000 pair-order accuracy, exceeding the validation-selected fair analytic estimator by 0.400 (base-world-clustered 95% interval [0.3767, 0.4217]). On a pinned public WAV MiniGrid split, paired evidence beats the strongest single-state control by 0.459–0.476 dynamic accuracy across five state complexities, while the released SparseIDM scores 0.940–0.953. The same audit rejects learned-method headroom in an RGB-D stress test because fair centroid velocity reaches 1.000 pair-order in all three families. It also rejects alternate chunks in public real-UR5 logs because they lack matched physical re-execution, despite a 0.950 standard-baseline test score. The contribution is an auditable admit-or-reject procedure, not a claim of general visual reasoning, real-robot transfer, or safety.

I. INTRODUCTION

Robot learning systems are increasingly evaluated on tasks described as dynamic, from broad simulated manipulation suites to visual real-robot control [1], [2], [3]. Yet a dynamic label need not require a dynamic decision rule. If current geometry, an action index, or an unordered set of observations predicts success, a verifier can score highly without using temporal direction. Matching only the final observation is also insufficient: a method may exploit aggregate displacement or the last two frames.

This distinction matters for candidate-action verification. A verifier is often used as a critic, reranker, or safety-adjacent filter, so benchmark performance is easily interpreted as evidence that it understands how a scene will evolve. Recent systems explicitly use interaction history or spatial reasoning to select among generated robot actions [4], [5]. That interpretation is warranted only when simpler observable routes have been excluded and outcomes remain tied to the candidate action. Benchmark validity and benchmark hardness are separate questions: a task can have clean splits and still be saturated by a fair analytic control.

We treat this as a benchmark-admission problem. The unit of intervention is the interpretation licensed by a benchmark, not a new verifier prediction. An execution-grounded counterfactual audit constructs paired histories that retain

prescribed observable information while candidate executions yield different outcomes. A progressive ladder tests static, action-template, order-invariant, final-two-frame, and candidate-degeneracy shortcuts. The result is deliberately asymmetric: passing can authorize a dynamics comparison, while any failed gate rejects only the corresponding claim. In particular, a learned-method claim is not authorized when a simple fair estimator closes the intended headroom.

Our contributions are:

- a three-gate admission intervention—integrity, temporal necessity, and method headroom—with an explicit observable contract, progressive counterfactual ladder, and refusal rule for unsupported method claims;
- the controlled TIMEARROW identification test: 92,000 execution-derived labels, an exact early-history pair swap, action-diverse C5/C10/C20 ranking, grouped splits, three OOD axes, leakage audits, and clustered uncertainty;
- a preregistered public WAV MiniGrid audit that confirms transition necessity across five state complexities without claiming a new method;
- two diagnostic rejections: a matched RGB-D construction whose fair geometric control saturates pair ordering, and public real-UR5 logs whose alternate chunks lack matched physical re-execution.

The central contribution is the evaluation intervention rather than the GRU. We deliberately bound every conclusion to the named procedural families and observations.

II. RELATED WORK

Large robot policies and simulation suites emphasize breadth of control and generalization [2], [1], [3]; our question is narrower and complementary: does an action verifier use the temporal evidence its benchmark is intended to test? Visual foresight and world-model methods predict action-conditioned futures for control [6], [7], whereas we neither learn a visual future model nor claim policy improvement.

History-aware verifiers rank candidates using past physical interactions [4], spatially grounded verifiers target geometric validity and goal progress [5], and World Action Verifier separates state plausibility from action reachability to improve world models [8]. These works design stronger verification mechanisms. We instead audit whether a controlled verifier benchmark identifies the evidence that its interpretation requires. We also apply our evidence controls to WAV’s public MiniGrid inverse-dynamics split; the result concerns that frozen benchmark only, not a competing verifier architecture or WAV’s robot results.

Shortcut learning describes decision rules that perform well for unintended reasons [9], [10]; causal confusion has also been documented in imitation learning [11]. Recent robot-policy studies connect shortcuts to dataset diversity and fragmentation [12], while LIBERO-CF counterfactually varies language intent under plausible visual layouts [13]. Our intervention instead matches candidate actions, current state, and the final history suffix while deriving differing outcomes from execution. Invariance and counterfactual methods motivate controlled comparisons across environments [14], [15]. Temporal-order verification and arrow-of-time recognition learn video representations from ordering cues [16], [17]; unlike them, we ask whether an action-conditioned evaluation actually requires those cues. PHYRE evaluates physical reasoning in controlled environments [18], CausalWorld exposes controllable factors in manipulation [19], and ManiSkill supplies GPU-parallel simulation and rendering [20]. We contribute an admission audit specialized to candidate-action verification: observable matching, execution-derived outcomes, action-diversity gates, and fair-estimator saturation checks. The public ASU TableTop data provide real UR5 language-conditioned manipulation trajectories [21]; we use them only to test whether a logged formulation qualifies for stronger causal interpretation.

III. BENCHMARK ADMISSION AS AN INTERVENTION

A. Verification task

For world w , let $H = (o_1, \dots, o_L)$ be an ordered observable history, A_w a candidate set, and $a \in A_w$ a candidate action. Executing a in the simulator produces $Y_w(a) \in \{0, 1\}$ under one common success rule. A verifier produces a score $s(H, a)$ used both to order a matched positive and negative example and to rank candidates within a world.

Fair inputs may contain timestamped observations, visibility or confidence, the robot TCP state, and the candidate action. They exclude family, mechanism, branch, object identity, future trajectory, and outcome fields. A counterfactual group contains members whose frozen observable constraints match but whose execution outcomes differ. Pair-order accuracy gives one for correct strict ordering, zero for reversed ordering, and the frozen tie value otherwise. Candidate Top-1 evaluates $Y_w(\arg \max_a s(H, a))$; it is a different estimand and is always reported separately.

B. Progressive shortcut ladder

Figure 1 summarizes the audit. Stage I measures shortcut-heavy ordinary data. Stage II matches current geometry and candidate fields. Stage III additionally matches unordered history statistics while changing signed temporal direction. Stage IV matches the final two frames and changes only earlier branch-identifying history. Stage V requires action-diverse candidate sets and independent OOD cells.

For six-frame histories, our final observable swap is

$$T(H) = (\text{reverse}(H_{1:4}), H_{5:6}). \quad (1)$$

It maps each generated partner exactly while retaining the two common synchronization frames. A generic reversal is

TABLE I
OBSERVABLE CONTRACT AND CORRESPONDING REJECTION TEST.

Threat	Preserved or matched	Rejection test
Current-state cue	o_6, a, TCP	Static/current model
Action template	Same a across pair	Action-only model
Orderless motion	Early-history multiset	Unordered model
Short suffix	Exact $o_{5:6}$	Final-two estimator
Candidate degeneracy	Mixed C5/C10/C20	Top-1 + diversity
Hidden identity	Grouped worlds, whitelist	Leakage audit
Analytic saturation	All legal geometry	Strongest fair rule

not a valid intervention because it moves these frames and changes timestamp alignment; we retain it only as a diagnostic. The pair-swap test evaluates $s(T(H), a)$ against the known partner, and score-swap consistency checks whether paired scores exchange as intended.

Table I makes the tested threat model explicit. Each row removes one route while retaining the information needed by later rows. This ordering prevents a chance static score from being mistaken for temporal necessity when a last-two-frame or geometric rule remains perfect.

C. Benchmark-admission rule

We distinguish three gates. *Validity* requires exact matching, grouped split containment, hidden-field exclusion, common label semantics, mixed candidate outcomes, and no candidate-index/template shortcut. *Temporal necessity* requires static, order-invariant, and final-two controls to fail on the intended pair distinction. *Method headroom* requires the strongest fair analytic or geometric estimator to remain below the relevant oracle or temporal comparator. Writing the gates as G_V, G_T, G_H , a temporal interpretation requires $G_V \wedge G_T$, whereas a learned-method comparison requires $G_V \wedge G_T \wedge G_H$. A failed gate is an audit result rather than a missing experiment; no gate provides a safety guarantee.

IV. CONTROLLED IDENTIFICATION BENCHMARK

A. Construction and labels

TIMEARROW is an identification test, not a surrogate for natural robot data. It contains damped moving capture, hysteretic moving container, and damped rotating-slot families with history-identifiable damping/drive and hysteretic mode memory. Each label is obtained by a GPU-PhysX proxy-to-target execution, using a common interception radius of 0.055 over execution steps 6–20. The analytic formulas used by fair baselines do not produce labels.

For transparency, the post-decision target motion is procedural. With branch $b \in \{0, 1\}$, sign $\sigma_b = (-1)^b$, and execution step n , the two frozen responses are

$$d_b^{\text{drv}}(n) = \sigma_b [0.025n + 0.020(1 - e^{-\lambda n}) \sin(\omega n + \pi b)], \quad (2)$$

$$d_b^{\text{hys}}(n) = 0.001\sigma_b [n - \delta_b]_+^2. \quad (3)$$

The three families map d_b to translation along y , translation along x at a fixed elevated goal, or angular motion on a 0.13-radius circle. At each simulator step the target pose is updated by this rule and the candidate moves a kinematic

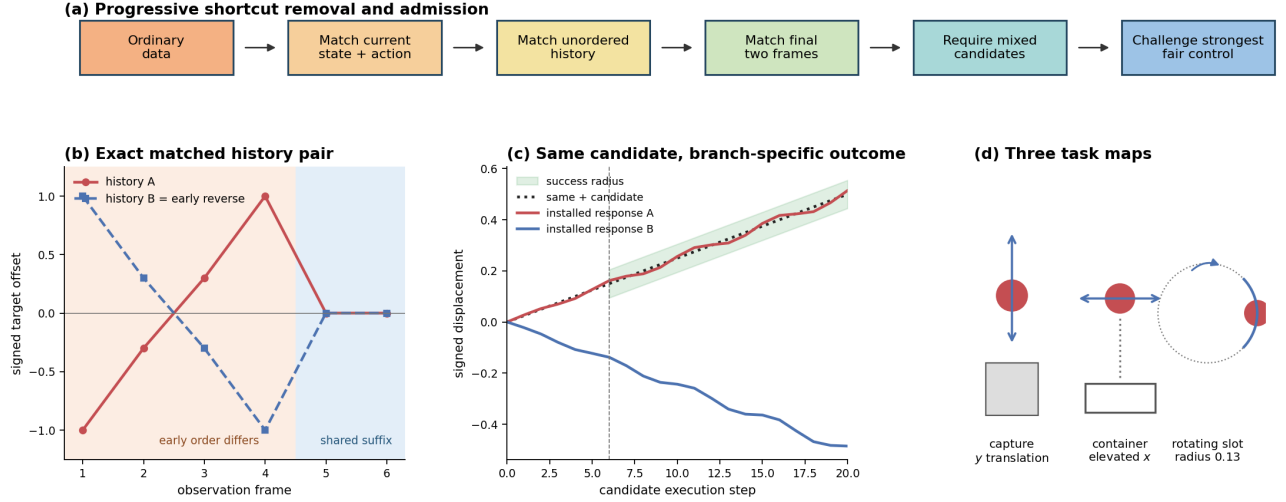


Fig. 1. Audit and state construction. (a) Each stage removes one named shortcut before testing headroom. (b) One history is the exact early-frame reverse of its partner and both share the final suffix. (c) An illustrative ID damping/drive world: the same positive-direction candidate falls within the 0.055 success band for one installed response but not the other after the step-6 evaluation onset. (d) The three task families map the response to distinct geometry.

TCP proxy; the recorded label is the monotone OR of the scene’s distance-based success output, never an analytic-baseline prediction.

The observation histories are also controlled rather than natural rollouts. Their first four target offsets are $[-s, -\rho s, \rho s, s]$ for one branch and the exact reverse for the other ($\rho = 0.30$ or 0.36); both then receive two identical synchronized decision poses. This makes the intended evidence and the exact intervention inspectable: earlier order identifies which frozen post-decision response is installed, while the current state and candidate are identical.

The final protocol uses five active family–mechanism ID cells, 300 ID worlds per cell, 100 worlds per active OOD cell, and aligned C5/C10/C20 candidate views. C20 contains ten candidates from each temporal direction; endpoint, pulse time, and amplitude vary by template. ID therefore contains 1,500 base worlds, two histories per world, 20 candidates per history, 60,000 labeled examples, and 30,000 matched pairs. The modulo-five world split yields 36,000/12,000/12,000 train/validation/test examples without dividing a world or pair. Across ID and independently generated OOD cells the frozen simulator budget is 92,000 candidate executions; this is execution accounting, not a claim of 92,000 independent worlds.

The physical OOD moves damping from $[0.12, 0.18]$ to $[0.26, 0.32]$ and drive frequency from $[1.15, 1.45]$ to $[1.85, 2.15]$. Delay OOD moves response onset from 0–1 to 2–3 steps. Held-mechanism OOD fits only damping/drive rows and tests fresh hysteretic-mode worlds. These axes are procedural and disjoint by construction; they do not approximate open-world or sim-to-real shift.

B. Comparators and metrics

Each state example exposes a six-by-20 observation history, timestamp, visibility and confidence per frame, a 20-by-

3 candidate trajectory, and the three-dimensional TCP state. Shortcut controls include static MLP, action-only, unordered-history MLP, and final-two finite difference. Fair analytic estimators include multi-frame and robust linear velocity, constant acceleration, exponential smoothing, and Kalman-style filters. Learned diagnostics include a lightweight GRU, temporal convolution, and ordered/unordered MLPs. Hidden state and oracle features are excluded from the fair comparison.

The selected GRU has 8,577 parameters: a 32-unit recurrent state followed by a 32-unit action-conditioned head. AdamW uses learning rate 0.015; the selection run uses three seeds and 100 full-batch updates. After validation-only model and analytic-baseline selection, the confirmation run uses five fixed seeds and 120 updates. Reported confirmation scores average logits across seeds.

The primary metric is pair-order accuracy. We additionally report Top-1, Top-3 success recall, utility-gain NDCG, normalized regret, swap controls, and inference latency. ID test contains 6,000 candidate-specific matched groups nested in 300 base worlds. The paired interval first averages the 20 candidate-pair differences within each world, then resamples the 300 worlds with 4,000 bootstrap draws (seed 7301). It characterizes variation over these frozen procedural worlds, not an open-world population. Ties receive 0.5 throughout.

V. STATE RESULTS

Validation selects GRU and MultiFrameLinearVelocity. Table II shows the frozen test comparison. GRU reaches 1.000 ID pair-order versus 0.600 for the fair analytic estimator, a +0.400 difference with base-world-clustered 95% interval $[0.3767, 0.4217]$. The advantage remains positive under physical-parameter, delay, and held-mechanism OOD; the weaker held-mechanism score (0.787) is reported rather than hidden. ID uses validation selection for both sides.

TABLE II

STATE PAIR-ORDER ACCURACY. THE ANALYTIC METHOD IS SELECTED ON VALIDATION FOR ID; EACH OOD ROW REPORTS ITS FROZEN BEST FAIR ANALYTIC COMPARATOR.

Split	GRU	Fair analytic	Δ
ID	1.000	0.600	+0.400
Physical OOD	1.000	0.500	+0.500
Delay OOD	1.000	0.857	+0.143
Held mechanism	0.787	0.667	+0.120

For each OOD row we instead report the strongest fair analytic rule in that test cell, an intentionally conservative oracle-control comparison rather than a deployable selection procedure.

Table III(a) reports every analytic and learned ID comparison, not only the winners. The three-seed ordered MLP reaches 0.900 and the selection-run GRU reaches 0.985 on test, showing that the signal is not unique to one confirmation run. Conversely, temporal convolution remains at 0.500; an architecture’s failure is not evidence that the cue is absent. Panel (b) exposes heterogeneity: the strongest analytic control saturates the rotating-slot/hysteresis cell at 1.000, while it is at 0.500 in the other four cells. Our aggregate claim therefore does not assert learned headroom in every family-mechanism cell.

Candidate selection groups candidates by family, mechanism, world, and branch: exactly one observable history. We explicitly reject the tempting but invalid key that pools both counterfactual branches. All 600 test histories at each candidate-set size contain both outcomes. Panel (c) uses independently saved raw scores: GRU Top-1 is 1.000 at C5/C10/C20, whereas static and action controls are 0.500. This history-key audit leaves pair-order and OOD values unchanged.

The controls in Table III(d) support the intended interpretation. Correct history is perfect while final-two-only is chance, so the result needs earlier observations. The exact pair swap and score-swap consistency are both 1.000. Applying the exact early swap while retaining original labels reverses the ordering completely (0.000); a generic full reversal remains at 0.982 and is therefore not a valid causal test. All C20 worlds are nondegenerate, every branch changes at least one candidate outcome, candidate-template success rates are balanced, and maximum candidate-index leakage is zero. C20 p95 inference latency is 0.816 ms on the recorded GPU. A red-team audit reports no blocking finding. Its one warning records a bookkeeping correction to the C20 diversity audit; no simulator, label, or score changed. The final history-key ranking correction is likewise versioned and retains all 14 score arrays for independent recomputation.

Perfect pair scores should be read as controlled diagnostic separation, not as evidence of universal physical reasoning. This is also why we next apply the admission audit to visual observations instead of immediately proposing a larger

verifier.

VI. BENCHMARK-ADMISSION STRESS TESTS

A. Public WAV MiniGrid transition check

We first apply the audit to an established public verifier benchmark. WAV releases a MiniGrid state-complexity split in which a model infers one of seven actions from (s_t, s_{t+1}) and carried-object state [8]. Its official dynamic-accuracy metric executes the predicted action with a public physics oracle and scores only channels that changed in the logged transition; it can therefore credit distinct actions with the same physical outcome. We pin upstream commit 527159b, its checkpoint, source, and six data-file hashes. The official o6 training set has 2,000 transitions; untouched o6 and o8/o10/o12/o14 tests contain 1,000 and 2,500 transitions, respectively.

Before training any control, we froze five seeds, a class-stratified 80/20 development split within o6, and three copies of one CNN architecture. The current-only and next-only cells duplicate one state into both network slots; the paired cell receives the unmodified transition. All use identical optimization and validation action-accuracy selection. We separately evaluate the released SparseIDM checkpoint over five inference seeds and a fixed transition rule using agent pose, the front cell, and carried-object changes. No test split selects a model or hyperparameter.

Table IV passes all preregistered gates. Paired evidence improves over the strongest single-state control by 0.469, 0.475, 0.476, 0.468, and 0.459 from o6 through o14. The released SparseIDM remains at 0.940–0.953, and the transition rule exceeds every single-state control. This is positive external evidence that the named inverse-dynamics benchmark requires the observed state transition under our controls. It does not audit WAV’s Action Following Score, its robot setting, or real-robot verification.

B. Visual RGB-D construction

We construct a separately versioned visual pilot because exact RGB-D reconstruction of the state histories cannot be established after the fact. For each of three procedural families, it contains 128 base worlds, two matched histories, ten shared candidate actions, six 128×128 RGB-D frames, and 2,560 candidate executions (7,680 total). Pair members share current RGB, depth, and action fields but differ in earlier history. Splits contain complete world groups; segmentation, identifiers, branch, mechanism, and future fields are excluded. The storage audit verifies exact current/final-two/action, unordered-history, mask, and metadata matching; no malformed group is found. Pair-label disagreement is 1.000 for moving cube/window and 0.974 for rotating window, and pair-order uses only opposite-label pairs. The fraction of histories with mixed C10 outcomes is 1.000, 1.000, and 0.977, respectively.

We use a held test camera and the union of ten non-complete conditions: random/burst missing frames, asynchronous RGB or depth, missing final frame, target occlusion, irregular timestamps, and latency. The frozen artifact

(a) Complete ID comparison				(b) Frozen ID test cells				
Category	Method	Val.	Test	Family	Mechanism	GRU	Strongest analytic	Score
Analytic	Last-two finite difference	0.500	0.500	Damped capture	Damping/drive	1.000	Last-two FD	0.500
Analytic	Multi-frame linear velocity	0.600	0.600	Damped capture	Hysteresis	1.000	Last-two FD	0.500
Analytic	Robust linear velocity	0.600	0.600	Rotating slot	Hysteresis	1.000	Multi-frame linear	1.000
Analytic	Constant-acceleration fit	0.400	0.400	Moving container	Damping/drive	1.000	Last-two FD	0.500
Analytic	Alpha-beta filter	0.600	0.600	Moving container	Hysteresis	1.000	Last-two FD	0.500
Analytic	Kalman constant velocity	0.600	0.600	(c) History-keyed candidate ranking				
Analytic	Kalman constant acceleration	0.600	0.600	Set	Contexts	Mixed	GRU Top-1	Static
Analytic	Exponential smoothing	0.400	0.400	C5	600	1.000	1.000	0.500
Learned (3)	Static MLP	0.500	0.500	C10	600	1.000	1.000	0.500
Learned (3)	Action-only MLP	0.500	0.500	C20	600	1.000	1.000	0.500
Learned (3)	Unordered-history MLP	0.500	0.500	(d) Validity and temporal-necessity controls				
Learned (3)	Temporal convolution	0.500	0.500	Audit	Pair-order / rate			
Learned (3)	Ordered temporal MLP	0.900	0.900	Correct history	1.000			
Learned (3)	GRU selection run	0.987	0.985	Final two only	0.500			
Confirm. (5)	GRU confirmation	1.000	1.000	Valid pair swap	1.000			
				Score-swap consistency	1.000			
				Nondegenerate C20 worlds	1.000			
				C20 dynamic Top-1	1.000			

TABLE III

STATE DIAGNOSTICS. PARENTHETICAL COUNTS IN (A) ARE SEEDS. CELL-WISE ANALYTIC RULES IN (B) ARE STRONGEST-IN-CELL STRESS CONTROLS, NOT TEST-SELECTED DEPLOYABLE MODELS. “MIXED” IN (C) IS THE FRACTION OF HISTORY CONTEXTS CONTAINING BOTH OUTCOMES. PANEL (D) REPORTS FINAL AUDIT GATES.

TABLE IV

EXTERNAL AUDIT OF THE PUBLIC WAV MiniGrid STATE-COMPLEXITY SPLIT. ENTRIES ARE OFFICIAL DYNAMIC ACCURACY; LEARNED CELLS AND SPARSEIDM REPORT MEAN±SAMPLE STANDARD DEVIATION OVER FIVE SEEDS. “STRONGEST SINGLE” IS THE LARGER OF THE SEPARATELY EVALUATED CURRENT-ONLY AND NEXT-ONLY MEANS AT EACH COMPLEXITY. THE TRANSITION RULE HAS 1.000 ACTION ACCURACY; ITS SUBUNIT DYNAMIC SCORE EXPOSES THE PUBLIC ORACLE/LOG MISMATCH AND IS TREATED AS AN OBSERVED CEILING, NOT USED TO NORMALIZE OTHER SCORES.

Objects	N	Strongest single	Paired CNN	SparseIDM	Transition rule
6	1000	0.410±0.024	0.879±0.012	0.947±0.000	0.947
8	2500	0.378±0.052	0.853±0.020	0.953±0.000	0.958
10	2500	0.329±0.047	0.805±0.044	0.940±0.001	0.952
12	2500	0.343±0.042	0.811±0.043	0.951±0.001	0.962
14	2500	0.341±0.083	0.800±0.041	0.947±0.002	0.969

calls this regime `combined_held_dropout`; here we use the more precise term *held-camera + corruption*. RGB-D and depth-backprojected world points are pooled deterministically from 128×128 to 16×16 . Fair controls include action-only, current and unordered observations, ordered RGB-D GRU, world-frame point-cloud GRU, and a segmentation-free MLP over per-frame global centroid, centroid difference, and speed. Calibration is legal input; target masks and the separate oracle files are not. All learned visual controls use seeds 17/29/43 and the same grouped fit split.

Table V gives three-seed point estimates. Action-only and current RGB-D remain exactly at 0.500. Ordered RGB-D GRU reaches only 0.537–0.665, while world-frame point-cloud normalization improves this to 0.727–0.860. However, centroid velocity reaches 1.000 pair-order in all three fam-

ilies, matching the clean-state oracle and leaving zero pair-order headroom. Because this control pools every valid scene point rather than using object identity, its success is itself a global-motion shortcut. The task therefore passes integrity and temporal-necessity checks but fails method admission.

This negative decision reveals a second evaluation trap. Pair ordering and C10 action selection are not interchangeable: although centroid pair-order is 1.000, its Top-1 success is only 0.653–0.727. A method paper could still study multi-candidate ranking under a newly frozen protocol, but it cannot cite the current paired task as unsolved learned-method headroom. The audit prevents a post-hoc method claim instead of treating a stopped experiment as missing evidence.

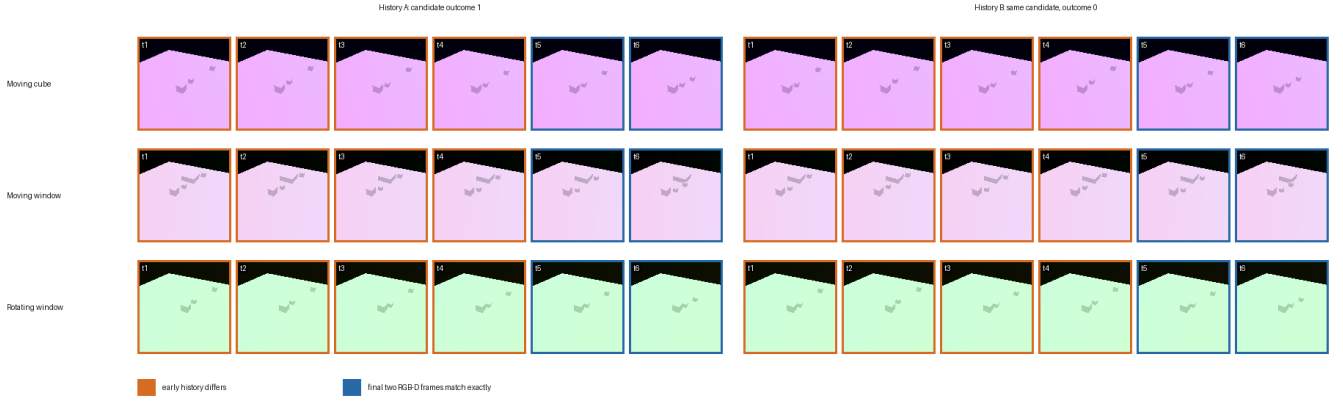


Fig. 2. Audited complete-history examples from one held-camera test world. Orange frames contain the differing early history; blue frames are the exact common RGB-D suffix. For candidate slot 0, history A’s execution succeeds and history B’s fails in every displayed family. The montage is generated from frozen bundles after checking nine exact source hashes; RGB is shown, while equality is asserted for both RGB and depth.

TABLE V

VISUAL STRESS TEST UNDER HELD-CAMERA + CORRUPTION (THE FROZEN `COMBINED_HELD_DROPOUT` KEY, ALL NON-COMplete TEST HISTORIES). COLUMNS EXCEPT “CENT. TOP-1” ARE PAIR-ORDER ACCURACY. THE FROZEN VISUAL CONFIDENCE INTERVALS ARE INTENTIONALLY SUPPRESSED BECAUSE THEIR SCORE- ENSEMBLE BOOTSTRAP ESTIMAND DOES NOT MATCH THE DISPLAYED SEED-AVERAGE POINT ESTIMATE.

Family	Action	Current	RGB-D	World-PC	Centroid	Cent. Top-1
Moving cube	0.500	0.500	0.665	0.860	1.000	0.673
Moving window	0.500	0.500	0.537	0.727	1.000	0.653
Rotating window	0.500	0.500	0.569	0.767	1.000	0.727

C. Public real-robot log scope check

We also apply the admission rule to a frozen formulation built from public ASU TableTop UR5 logs [21]. It contains 84 episodes from 11 official action–object tasks and 1,151 C5 candidate sets. The harder split trains on bread/bottle/cube, validates on Pepsi, and tests on unseen Coke/milk goal objects using whole episodes. Action-, task-, language-, source-, and shuffled-smoothness validation controls are at most 0.534, but a standard proprio-action GRU reaches 0.972 validation and 0.950 test pair-order.

This does not establish counterfactual real-robot verification. The positive is the actual next action chunk; the four negatives are distant, other-episode, different-task, or reversed chunks drawn from existing logs. They were never executed from the same decision state and therefore have no candidate-specific physical outcomes. The audit consequently rejects the formulation at both execution grounding and method headroom, before training a proposed verifier.

Table VI makes the asymmetric outcome explicit. The controlled state identification test and public WAV MiniGrid split pass their respective temporal-evidence gates. The visual construction passes integrity and temporal-necessity checks but loses all pair-order headroom to a fair geometric estimator; the real-log task lacks matched re-execution. These negative cases are useful evidence for the audit procedure precisely because they are rejected.

VII. DISCUSSION AND LIMITATIONS

The state construction is procedural and simulator-specific. It uses clean state observations, controlled TCP geometry, limited embodiment diversity, and designed counterfactual factors. Its histories are instantiated from the frozen response family rather than collected from unconstrained interaction; the positive result therefore establishes diagnostic separation, not natural dynamics fidelity. The visual stress test adds RGB-D, occlusion/corruption, camera shift, and geometry extraction, but it is also procedural and is explicitly rejected as a hard benchmark. Its backgrounds and global scene motion make centroid saturation possible; adding visual complexity without rerunning the admission ladder would not establish a fix. The controlled experiments do not include language or contact-rich robot outcomes. The public-log check adds language and real trajectories but still lacks alternate-action robot executions. None includes end-to-end policy learning or safety validation.

The external WAV check broadens benchmark provenance but not embodiment: it is a discrete MiniGrid inverse-dynamics task, not the WAV Action Following Score or robot experiment. Its public physics oracle does not exactly reproduce every logged changed channel even for the true action, so the transition rule’s 0.947–0.969 dynamic accuracy is an empirical ceiling rather than 1.000. This does not affect the preregistered paired-versus-single comparison, but it limits cross-metric interpretation.

Counterfactual language here is local to the frozen gen-

TABLE VI

BENCHMARK-ADMISSION DECISIONS GENERATED FROM THE FROZEN REPORTS. “INTEGRITY” COVERS EXACT MATCHING, GROUPED SPLITS, HIDDEN-FIELD EXCLUSION, AND CANDIDATE DIVERSITY WHERE APPLICABLE. STATE ADMISSION IS AN AGGREGATE CLAIM; ONE FAMILY-MECHANISM CELL IS ANALYTICALLY SATURATED. THE WAV ROW COVERS ONLY ITS MINIGRID INVERSE-DYNAMICS SPLIT, AND THE UR5 ROW IS LOGGED-CONTINUATION IDENTIFICATION, NOT MATCHED RE-EXECUTION.

Benchmark	Integrity	Temporal evidence	Fair-control test	Decision
State observations	Pass	Final-two 0.500	Analytic 0.600 vs GRU 1.000	Admit controlled comparison
Public WAV MiniGrid	Pinned upstream	Single ≤ 0.356	SparseIDM 0.948	Admit transition evidence
Visual RGB-D stress test	Pass	Final-two 0.500; unordered ≤ 0.535	Centroid 1.000	Reject method claim
Public UR5 logs	Logged groups pass	Trivial val. ≤ 0.534	Proprio GRU 0.972 val.	Reject: no re-execution

erator and exact swap; we do not claim an intervention on arbitrary real videos. Execution grounding separates the success label from analytic estimators, but does not by itself eliminate simulator artifacts. The world-clustered state interval and visual three-seed point estimates have different statistical units and are never pooled. Finally, a benchmark that defeats the named controls may admit an unknown shortcut; publishing the observable contract and admission tests is therefore part of the contribution. Perfect state scores may also indicate that the final procedural distinction is cleaner than natural dynamics. Future work should apply the same ladder to matched real-robot re-executions; analytic saturation would remain an informative rejection rather than missing evidence.

VIII. REPRODUCIBILITY AND CONCLUSION

A downstream-only builder reads sixteen frozen JSON reports, checks their exact SHA-256 hashes plus the corrected-ranking raw-score hash, enforces 36 scientific invariants, suppresses the invalid visual CI, and deterministically emits the paper tables and a claim ledger. No training or evaluator code is imported during table generation. The ranking correction stores all five GRU and six control-seed score vectors plus their means; a verify-only path recomputes every history-keyed metric. The visual montage independently checks nine bundle/candidate/label hashes and the exact RGB-D suffix. The package also records split and leakage audits, execution accounting, and both non-blocking bookkeeping fixes plus the world-clustered interval correction. The external audit stores confusion counts and prediction hashes for every method, seed, and complexity; an independent verifier recomputes all summaries and 15 frozen gates.

We conclude that dynamic labels and chance-level static controls are necessary but insufficient evidence of temporal reasoning. Execution-grounded matched pairs expose named shortcuts in the state setting, while the same admission logic confirms transition necessity on a pinned public verifier split and correctly rejects a visual construction saturated by simple geometry and a real-log task without matched alternate-action outcomes. This combination—positive controlled evidence and an explicit refusal to claim headroom when it is absent—makes action-verifier evaluation more falsifiable.

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REFERENCES

- [1] S. James, Z. Ma, D. R. Arrojo, and A. J. Davison, “Rlbench: The robot learning benchmark & learning environment,” *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3019–3026, 2020.
- [2] A. Brohan *et al.*, “Rt-1: Robotics transformer for real-world control at scale,” in *Robotics: Science and Systems XIX*, 2023.
- [3] S. Nasiriany, A. Maddukuri, L. Zhang, A. Parikh, A. Lo, A. Joshi, A. Mandlekar, and Y. Zhu, “Robocasa: Large-scale simulation of everyday tasks for generalist robots,” in *Robotics: Science and Systems*, 2024.
- [4] Y. Li, X. Mao, Y. Yuan, K. Sim, B. Eisner, and D. Held, “Learn from what we have: History-aware verifier that reasons about past interactions online,” in *Proceedings of the 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305, 2025, pp. 2063–2092.
- [5] G. Zhao, L. Guo, J. Zhu, J. Fu, Y. Mei, B. Cao, J. Jiang, X. He, and J. Liu, “Verispace: Spatially grounded action verification for vision-language-action models,” *arXiv preprint arXiv:2606.10568*, 2026.
- [6] F. Ebert, C. Finn, S. Dasari, A. Xie, A. Lee, and S. Levine, “Visual foresight: Model-based deep reinforcement learning for vision-based robotic control,” in *IEEE International Conference on Robotics and Automation*, 2018.
- [7] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, “Mastering diverse control tasks through world models,” *Nature*, vol. 640, pp. 647–653, 2025.
- [8] Y. Liu, F. Feng, L. Kong, W. Lu, J. Tang, K. Zhang, K. Murphy, C. Finn, and Y. Du, “World action verifier: Self-improving world models via forward-inverse asymmetry,” *arXiv preprint arXiv:2604.01985*, 2026.
- [9] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.
- [10] S. Lapuschkin, S. W. Aldchen, A. Binder, G. Montavon, W. Samek, and K.-R. Müller, “Unmasking clever hans predictors and assessing what machines really learn,” *Nature Communications*, vol. 10, no. 1, p. 1096, 2019.
- [11] P. de Haan, D. Jayaraman, and S. Levine, “Causal confusion in imitation learning,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [12] Y. Xing, X. Luo, J. Xie, L. Gao, H. T. Shen, and J. Song, “Shortcut learning in generalist robot policies: The role of dataset diversity and fragmentation,” in *Proceedings of the 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305, 2025, pp. 3239–3266.
- [13] Y. Fang, Y. Feng, D. Jing, J. Liu, Y. Yang, Z. Wei, D. Szafir, and M. Ding, “When vision overrides language: Evaluating and mitigating counterfactual failures in vision-language-action models,” *arXiv preprint arXiv:2602.17659*, 2026.
- [14] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz, “Invariant risk minimization,” *arXiv preprint arXiv:1907.02893*, 2019.

- [15] S. Wachter, B. Mittelstadt, and C. Russell, “Counterfactual explanations without opening the black box: Automated decisions and the gdpr,” *Harvard Journal of Law & Technology*, vol. 31, no. 2, 2017.
- [16] I. Misra, C. L. Zitnick, and M. Hebert, “Shuffle and learn: Unsupervised learning using temporal order verification,” in *European Conference on Computer Vision*, 2016, pp. 527–544.
- [17] D. Wei, J. J. Lim, A. Zisserman, and W. T. Freeman, “Learning and using the arrow of time,” in *IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 8052–8060.
- [18] A. Bakhtin, L. van der Maaten, J. Johnson, L. Gustafson, and R. Girshick, “PHYRE: A new benchmark for physical reasoning,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [19] O. Ahmed, F. Träuble, A. Goyal, A. Neitz, Y. Bengio, B. Schölkopf, M. Wüthrich, and S. Bauer, “Causalworld: A robotic manipulation benchmark for causal structure and transfer learning,” in *International Conference on Learning Representations*, 2021.
- [20] S. Tao *et al.*, “Maniskill3: Gpu parallelized robotics simulation and rendering for generalizable embodied ai,” *arXiv preprint arXiv:2410.00425*, 2024.
- [21] Y. Zhou, S. Sonawani, M. Phielipp, S. Stepputtis, and H. Amor, “Modularity through attention: Efficient training and transfer of language-conditioned policies for robot manipulation,” in *Proceedings of the 6th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 205, 2023, pp. 1684–1695.